3	(a)	(i)	$pV = (1/3)\ N\ m\langle c^2\rangle$ where $\rho = Nm/v$ $p = 1/3\ \rho\ \langle c^2\rangle$ $(\langle c^2\rangle)^{0.5} = [(3 \times 2.1 \times 10^5) / 1.5]^{0.5}$ $= 648\ (650)\ \text{m s}^{-1}$	 1 1
		(ii)	$½\ m\ \langle c^2\rangle = 3/2\ kT$, $m = Mr / N_A$ $T = (m\ \langle c^2\rangle) / 3k$ $= [(14 \times 10^{-3}) / (6.02 \times 10^{23})] * (648^2) / (3 \times 1.38 \times 10^{-23})$ $= 236\ (240)$ K	 1 1
		(iii)	When the temperature of the gas is increased, the <u>frequency of collisions on the wall</u> will increase and the <u>root-mean-square speed of the molecules will increase.</u> The gas molecules exert <u>a greater force on the walls</u> of the container due to greater change in momentum. Hence <u>pressure will increase</u>.	1 1 1
	(b)		Energy to be removed by refrigerator = energy loss by chicken + air $Q = 2.0 \times 3.48 \times 10^3\ (25 – 4.0) + 1.25 \times 1.50 \times (1.02 \times 10^3) \times (25 – 4.0)$ $= 1.86 \times 10^5$ J $Qsup = (1.86 \text{ x } 10^5) / 0.80$ $t = Qsup / P = 1160\ (1200)$ s	 1 1 1

4	(a)	(i)	$45(2\pi)$	